

TrellisBsP: Trellis-Based Belief-Selective Propagation for MIMO Detection

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Abstract—Compared to its linear counterparts, the belief propagation (BP) MIMO detector achieves better performance in challenging MIMO scenarios. For practical applications, the recently emerged belief-selective propagation (BsP) detector mitigates the error floor issue and reduces the complexity. However, it still suffers from intractable complexity for high-order MIMO scenarios, and the serial message-updating mechanism limits its hardware feasibility. In this letter, a trellis-based BsP (TrellisBsP) detection algorithm is proposed. Two unexplored approaches: *trellis representation* and *pathfinding strategy* are utilized to enable parallel message-updating while achieving complexity reduction. Benchmarking with the BsP detection, the proposed TrellisBsP saves up to 72.76% complexity in an 8×4 , 256-QAM MIMO system without error performance penalty. In addition, the proposed TrellisBsP has great potential in hardware implementation due to its parallel message-updating mechanism.

Index Terms—Multiple-input multiple-output (MIMO) detection, belief-selective propagation (BsP), trellis representation, pathfinding strategy.

I. INTRODUCTION

UBIQUITOUS communications places increasingly stringent demand on wireless technology. To achieve the ambitious objectives of the forthcoming 6G applications [1], multiple-input multiple-output (MIMO), which sufficiently enlarges both spectrum efficiency (SE) and energy efficiency (EE), evolves into a large scale with hundreds or thousands of antennas. However, this advancement imposes greater challenges on signal detection algorithms.

The *maximum a posteriori* (MAP) detection offers optimal performance but at the cost of an exponentially increasing complexity. Consequently, the linear minimum mean square error (LMMSE) [2] detector is the most commonly utilized in MIMO systems due to its ultra-low computational complexity. However, its bottleneck mainly lies in poor performance, especially in challenging large MIMO scenarios. Alongside the linear detectors, there are many well-established message-passing detectors such as the large-MIMO approximate message passing algorithm (LAMA) [3], [4],

expectation propagation (EP) [5], [6], and belief propagation (BP) [7], [8], [9], which can deliver better performance at the expense of higher complexity. Due to its near-optimal error performance and implementation-friendly hardware architecture, BP detection has garnered wide attention in recent years. The original BP detection and its approximate versions are introduced in [7], [8], among which the real-domain *Gaussian* approximate of interference BP (RD-GAI-BP) is the most common-used one. The more recent belief-selective propagation (BsP) [9] together with its hardware implementation-aware version [10] has shown great potential due to its performance and complexity merits. However, the BsP detection still suffers from two problems when applied to realistic MIMO scenarios: *i*) its message-updating complexity is intractable for high-order modulated MIMO systems; *ii*) its serial message-updating mechanism results in low hardware efficiency. These limitations motivate us to think of an approach that cannot only mitigate the computational complexity but also improve the hardware potential for the BsP detection.

This letter proposes a revised BsP detection algorithm based on trellis representation for each FN (named TrellisBsP). Two unexplored methods: *trellis representation* and *pathfinding strategy* are utilized to lower the computational complexity and enable parallelism for the message updating. More specifically, each FN begins the message update process by constructing a message trellis, where the incoming messages are rearranged following an order of “high-to-low reliability”. Then, the pathfinding strategy is utilized to establish a path set that involves the constellation points with relatively large reliability. Finally, the output messages are computed in parallel using the selective paths. Since the proposed TrellisBsP detection only updates the messages corresponding to constellation points with high beliefs for each path, it achieves lower complexity compared with the BsP detection. To the author’s best knowledge, the proposed TrellisBsP is the very first BP detection that implements FN’s parallel message updating. Numerical results showcase that the proposed TrellisBsP detection achieves up to 72.76% complexity reduction compared with the BsP detection in an 8×4 256-QAM MIMO system without error performance penalty. Compared with the well-established expectation propagation (EP) [5] and LAMA [3] detection, the proposed TrellisBsP saves 65.57% and 93.98% complexity in the 8×4 256-QAM MIMO scenario. It is noted that the trellis representation originates from the trellis extended Min-Sum (T-EMS) decoding algorithm for non-binary low-density parity-check (NB-LDPC) codes [11]. However, compared to the T-EMS decoding, our proposed TrellisBsP detection has no check equation constraint. Therefore, it builds

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the trellis without involving the *extra column* and domain conversion.

The rest of this letter is organized as follows. The background is introduced in Section II. The technique details and numerical results are provided in Sections III and IV. Section V concludes this letter.

II. PRELIMINARIES

A. MIMO System Model

Consider an uplink flat-fading massive MIMO system equipped with N_r receive antennas and N_t transmit antennas. Denote the transmitted (TX) symbol vector as $\tilde{\mathbf{s}} = [\tilde{s}_1, \dots, \tilde{s}_{N_t}]^T \in \mathbb{C}^{N_t \times 1}$ with unitary average power for each component. Assuming that all TX symbols are selected from the $\mathcal{Q}$ -QAM constellation set and its cardinality is $|\mathcal{Q}| = 2^M$, where M represents the modulation order. The received (RX) symbol vector is denoted as $\tilde{\mathbf{y}} = [\tilde{y}_1, \dots, \tilde{y}_{N_r}]^T \in \mathbb{C}^{N_r \times 1}$, given by

$$\tilde{\mathbf{y}} = \tilde{\mathbf{H}}\tilde{\mathbf{s}} + \tilde{\mathbf{n}}, \quad (1)$$

where $\tilde{\mathbf{H}} = [\tilde{\mathbf{h}}_1^T, \dots, \tilde{\mathbf{h}}_{N_r}^T]^T \in \mathbb{C}^{N_r \times N_t}$ stands for a complex channel matrix, $\tilde{\mathbf{h}}_i$ denotes the i -th row of $\tilde{\mathbf{H}}$, and each coefficient $\tilde{h}_{ij} \in \tilde{\mathbf{H}}$ is an independent and identically distributed (i.i.d.) complex following zero-mean and unitary variance, i.e., $\tilde{h}_{ij} \sim \mathcal{CN}(0, 1)$. And $\tilde{\mathbf{n}} \in \mathbb{C}^{N_r \times 1}$ is a complex Gaussian noise vector with zero-mean and variance σ_n^2 . In this letter, it is assumed that the channel state information (CSI) is perfectly known on the receiver side.

Generally, *real value decomposition* (RVD) is utilized for MIMO detections. In this case, (1) can be equivalently converted into real-domain expression, rewritten as:

$$\mathbf{y} = \mathbf{H}\mathbf{s} + \mathbf{n}, \quad (2)$$

where $\mathbf{y} = \begin{bmatrix} \Re(\tilde{\mathbf{y}}) \\ \Im(\tilde{\mathbf{y}}) \end{bmatrix}$, $\mathbf{s} = \begin{bmatrix} \Re(\tilde{\mathbf{s}}) \\ \Im(\tilde{\mathbf{s}}) \end{bmatrix}$, $\mathbf{n} = \begin{bmatrix} \Re(\tilde{\mathbf{n}}) \\ \Im(\tilde{\mathbf{n}}) \end{bmatrix}$, and the real-domain channel matrix can be expressed as

$$\mathbf{H} = \begin{bmatrix} \Re(\tilde{\mathbf{H}}) & -\Im(\tilde{\mathbf{H}}) \\ \Im(\tilde{\mathbf{H}}) & \Re(\tilde{\mathbf{H}}) \end{bmatrix} \in \mathbb{R}^{2N_r \times 2N_t}. \quad (3)$$

The operations $\Re(\cdot)$ and $\Im(\cdot)$ represent the real and imaginary components of $(\cdot)$, respectively. After this real-value conversion, TX symbols are all chosen from the real-value constellation set $\mathcal{Q}$, and its size is $|\mathcal{Q}| = \sqrt{|\tilde{\mathcal{Q}}|}$.

B. Belief-Selective Propagation Detection

Fig. 1 illustrates the factor graph model of the BsP detection, consisting of $2N_r$ FNs and $2N_t$ symbol nodes (SNs), denoted by f_i and S_j , respectively. The soft messages sent from f_i to S_j are denoted by β_{ij} and computed as

$$\beta_{ij}(k) = \max_{\substack{\mathbf{s}: s_j = \mu_k, \\ \mathbf{s}_{k/j} \in \mathcal{B}(d_m, d_f)}} \left\{ -\frac{1}{2\sigma_n^2} \|y_i - \mathbf{h}_i \mathbf{s}\|^2 + \sum_{t=1, t \neq j}^{2N_t} \alpha_{ti} \right\} \\ - \max_{\substack{\mathbf{s}: s_j = \mu_1, \\ \mathbf{s}_{k/j} \in \mathcal{B}(d_m, d_f)}} \left\{ -\frac{1}{2\sigma_n^2} \|y_i - \mathbf{h}_i \mathbf{s}\|^2 + \sum_{t=1, t \neq j}^{2N_t} \alpha_{ti} \right\}, \quad (4)$$

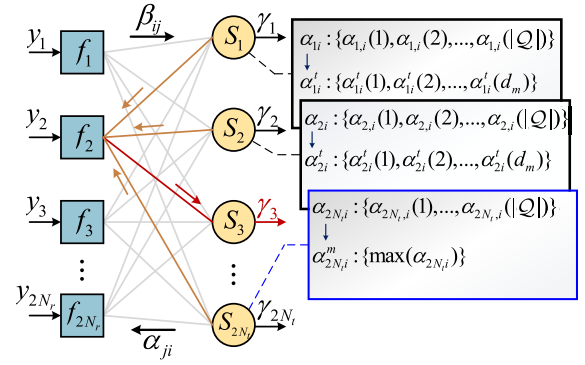


Fig. 1. Factor graph model for the BsP detection.

where μ_k represents the k -th constellation point, $\mathbf{s}_{k/j}$ denotes a TX vector without the j -th TX symbol. $\mathcal{B}(d_m, d_f)$ is the configuration set, wherein $d_f - 1$ *chosen edges* retain d_m largest soft messages (depicted as black boxes) and $2N_t - d_f$ *simplified edges* only keep the largest soft messages (depicted as blue boxes) to calculate β_{ij} . The α_{ji} are soft messages reversely sent from S_j to f_i , computed as

$$\alpha_{ji}(k) = \sum_{t=1, t \neq i}^{2N_r} \beta_{tj}(k). \quad (5)$$

The β and α messages are iteratively updated until the maximum number of iterations (denoted as Q_L) is reached. Then, the BsP detection will output soft messages for each TX symbol (denoted as γ_j) by

$$\gamma_j(k) = \sum_{i=1}^{2N_r} \beta_{ij}(k). \quad (6)$$

More details of the BsP detection can be found in [9].

III. THE PROPOSED TRELLISBSP DETECTION

A. Trellis Representation

In our proposed TrellisBsP detection, we build a sorted trellis to visually demonstrate the input *a priori* messages to each FN. We denote the trellis built for the i -th FN as $\mathcal{T}_i$. As illustrated in Fig. 2(a), each trellis is composed of $|Q|$ rows and $2N_t$ columns of constellation points. The constellation point at the m -th row and j -th column is denoted as $\mu_{c_i^j}^m$ and its corresponding *a priori* message is $\alpha_{c_i^j}^m$, combined to $(\mu_{c_i^j}^m, \alpha_{c_i^j}^m)$ as a tuple. Therefore, the i -th trellis can be expressed as $\mathcal{T}_i = (\mathcal{M}_i, \mathcal{A}_i)$, where $\mathcal{M}_i$ and $\mathcal{A}_i$ represent the set of constellation points and corresponding α messages, respectively. The constellation points of each column are sorted according to their α messages so that $\alpha_{c_i^j}^1 > \dots > \alpha_{c_i^j}^{|Q|}$. For each column, only the n_m most reliable constellation points are utilized for message updating. However, it is observed from Eq. (4) that constellation point μ_1 is inevitable for computing β messages. Therefore, μ_1 together with its α message, denoted as $(\mu_{c_i^*}^j, \alpha_{c_i^*}^j)$, is also applied for β message updating. We define $\hat{\mathcal{T}}_i^j$ as the subset involving all of the tuples utilized for message updates in the j -th column of $\mathcal{T}_i$, which is expressed as

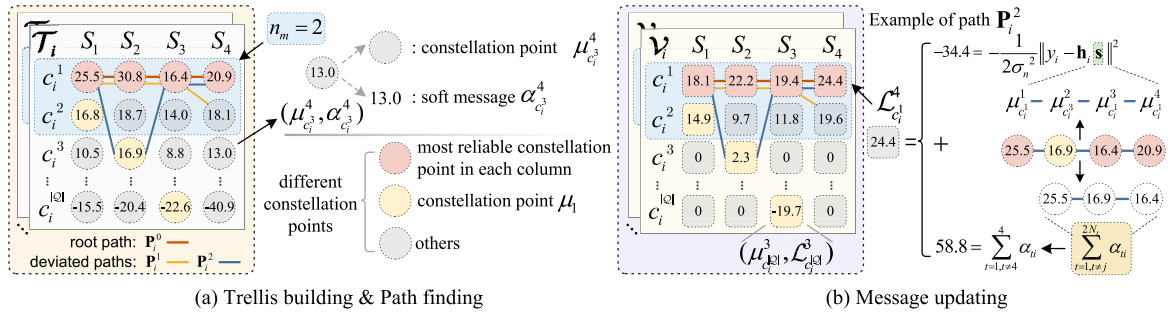


Fig. 2. Trellis representation for the proposed TrellisBsP with (a) Trellis building & Path finding and (b) Message updating. Each circle in (a) represents a tuple associated with *a priori* messages and each rounded square in (b) denotes a tuple corresponding to posterior messages.

$$\mathcal{T}_i \supseteq \left\{ \bigcup_{j=1}^{2N_t} \hat{\mathcal{T}}_i^j \right\}, \hat{\mathcal{T}}_i^j = (\hat{\mathcal{M}}_i^j, \hat{\mathcal{A}}_i^j) = [(\mu_{c_i^1}^j, \alpha_{c_i^1}^j), (\mu_{c_i^2}^j, \alpha_{c_i^2}^j), \dots, (\mu_{c_i^{n_m}}^j, \alpha_{c_i^{n_m}}^j)] \cup \{(\mu_{c_i^*}^j, \alpha_{c_i^*}^j)\}.$$

B. Pathfinding Strategy

The motivation of the proposed pathfinding strategy is to build up path sets with the selective constellation points and then compute β messages accordingly. As shown in Fig. 2(a), it is observed that each path involves $2N_t$ constellation points corresponding to $2N_t$ TX symbols. We define the path involving constellation points in the first row as the *root path*, denoted by $\mathbf{P}_i^0$, i.e., $\mathbf{P}_i^0 = [\mu_{c_i^1}^1, \dots, \mu_{c_i^{2N_t}}^1]$. Then, the path set is denoted as

$$\mathcal{P}_i(n_m, n_c) = \left\{ \mathbf{P}_i^k = \left[\mu_{c_i^1}^k, \mu_{c_i^2}^k, \dots, \mu_{c_i^{2N_t}}^k \right] : \forall \mu_{c_i^{k_j}}^j \in \hat{\mathcal{M}}_i^j, D[\mathbf{P}_i^0, \mathbf{P}_i^k] = n_c \right\}, \quad (8)$$

where $D[\mathbf{P}_i^0, \mathbf{P}_i^k] = n_c$ is a constraint that ensures the number of deviated constellation points between $\mathbf{P}_i^k$ and the root path is equal to n_c . We denote the number of paths in $\mathcal{P}_i(n_m, n_c)$ as N_p . Note that N_p is not a fixed number, but it has an upper bound of $N_p = \binom{2N_t}{n_c} n_m^{n_c}$, corresponding to the worst case that μ_1 in each column is out of the first n_m rows. For the reader's better understanding, a toy example of the trellis representation and pathfinding strategy is shown as follows.

Example 1: In Fig. 2(a), a detailed trellis with $|\mathcal{Q}|$ rows and 4 columns is built for f_i . Each column arranges the constellation points in descending order in terms of their α messages. The most reliable constellation point in each column is highlighted in pink while those circles indicating μ_1 are distinguished by yellow. The blue dashed rectangle involves the n_m most reliable constellation points in each column, with $n_m = 2$ in this example. With the built trellis, the path set $\mathcal{P}_i(2, 1)$ is derived. Three distinct paths are depicted for instance. First, the root path $\mathbf{P}_i^0$ involves the most reliable constellation point for each column. Then, two deviated paths: $\mathbf{P}_i^1$ and $\mathbf{P}_i^2$ are established with the constraint of $n_c = 1$. It is noted that there are other possible paths satisfying the constraint, but we don't draw them for the sake of simplicity.

Finally, all of the paths are incorporated into the set $\mathcal{P}_i(2, 1)$, utilized for β message updating accordingly.

C. Message Updating

With the aforementioned path sets, we can compute the posterior messages in parallel and fill them into a new trellis (denoted as $\mathcal{V}_i = (\mathcal{M}_i, \mathcal{L}_i)$), shown as Fig. 2(b). The structure of $\mathcal{V}_i$ shares similarities with the $\mathcal{T}_i$. However, each constellation point in $\mathcal{V}_i$ is paired with its corresponding posterior message to form a tuple, denoted as $(\mu_{c_i^m}^j, \mathcal{L}_{c_i^m}^j)$, instead of its input message $\alpha_{c_i^m}^j$. For those constellation points out of the chosen paths, we just maintain their posterior messages as zero. With the path sets, the posterior message $\mathcal{L}_{c_i^m}^j$ is computed as

$$\mathcal{L}_{c_i^m}^j = \max_{\substack{\mathbf{s} \in \mathbf{P}_i^0 \cup \mathcal{P}_i(n_m, n_c) \\ \mathbf{s}: s_j = \mu_{c_i^m}^j}} \left\{ -\frac{1}{2\sigma_n^2} \|y_i - \mathbf{h}_i \mathbf{s}\|^2 + \sum_{k=1, k \neq j}^{2N_t} \alpha_{c_i^k} \right\}. \quad (9)$$

According to (4), the related β messages are computed as

$$\beta_{ij}^{(l)}(\mu_{c_i^m}^j) = \mathcal{L}_{c_i^m}^j(l) - \mathcal{L}_{c_i^*}^j(l), \quad (10)$$

where l denotes the l -th iteration and $\mathcal{L}_{c_i^*}^j(l)$ represents the posterior message associated with μ_1 . Noted that the message computation for each path in (9) is independent, thus supporting parallel updating for all paths. If a constellation point is overlapped by multiple paths, its output messages will be continuously updated until reaching the maximum value. For clarity, we also provided an example to show how to compute the posterior message $\mathcal{L}_{c_i^m}^j$ in (9).

Example (cont'd): Fig. 2(b) provides an example of message updating for $\mathcal{L}_{c_i^1}^4$ with the path $\mathbf{P}_i^2$. According to (9), the message calculation involves two parts: likelihood probability and a priori probability. The former is computed with the constellation points in the blue path and the latter is derived from the α messages of incoming edges. Here we compute the likelihood probability as -34.4 and the *a priori* message component as 58.8 . As a result, the posterior message $\mathcal{L}_{c_i^1}^4$ is filled with 24.4 .

D. TrellisBsP Detection

The proposed TrellisBsP detection is summarized as Alg. 1. Specifically, before starting message updates, the initial α messages are provided by an LMMSE pre-processor. More

Algorithm 1: TrellisBsP Detection**Input** : received signal $\mathbf{y}$, channel matrix $\mathbf{H}$ **Output**: symbol LLRs $\boldsymbol{\gamma}$

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1 Initialization:  $\boldsymbol{\alpha} \leftarrow \text{InitMMSE}(\mathbf{y}, \mathbf{H})$ ;
2 for  $\text{Iter} \leftarrow 1$  to  $Q_L$  do
3   for  $i \leftarrow 1$  to  $2N_r$  do // update  $\boldsymbol{\beta}$  for each FN
4     build  $\mathcal{P}_i(n_m, n_c) \leftarrow \boldsymbol{\alpha}$ ;
5     par for  $k \leftarrow 1$  to  $N_p$  do // parallel loops
6        $\mathbf{s} \leftarrow \mathbf{P}_i^k \in \mathbf{P}_i^0 \cup \mathcal{P}_i(n_m, n_c)$ ,
7        $\Psi_i(\mathbf{s}) = -\frac{1}{2\sigma_n^2} \|\mathbf{y}_i - \mathbf{h}_i \mathbf{s}\|^2$ ;
8        $\Lambda_k = \sum_{j=1}^{2N_t} \alpha_{c_j}^j$ ;
9       for  $j \leftarrow 1$  to  $2N_t$  do
10         $\mathcal{L}^\dagger = \Psi_i(\mathbf{s}) + \Lambda_k - \alpha_{c_j}^j$ ;
11         $\mathcal{L}_{c_i}^j = \max\{\mathcal{L}_{c_i}^j, \mathcal{L}^\dagger\}$ ;
12       $\forall j, m, \mu_{c_i}^j \in \mathbf{P}_i^0 \cup \mathcal{P}_i(n_m, n_c)$ :
13       $\beta_{ij}(\mu_{c_i}^j) = \mathcal{L}_{c_i}^j - \mathcal{L}_{c_i}^*$ ;
14   for  $j \leftarrow 1$  to  $2N_t$  do // update  $\boldsymbol{\alpha}$  messages
15     update  $\gamma_j$  as (6);
16     for  $i \leftarrow 1$  to  $2N_r$  do
17        $\alpha_{ji} \leftarrow \gamma_j - \beta_{ij}$  as (5);

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details about the initialization can be found in [10]. During each iteration, first, path sets are constructed based on the trellis according to incoming $\boldsymbol{\alpha}$ messages for every FN. Second, for different paths, posterior messages and $\boldsymbol{\beta}$ messages are updated according to (9) and (10). Finally, the $\boldsymbol{\alpha}$ and $\boldsymbol{\gamma}$ messages are updated as (5) and (6) for each SN, respectively. When the maximum number of iterations is achieved, the TrellisBsP will stop the message updates and output symbol LLRs ($\boldsymbol{\gamma}$ messages).

IV. NUMERICAL RESULTS

In this section, numerical results of the provided TrellisBsP are provided, compared with the LMMSE [2], the RD-GAI-BP [8], LAMA [3], the BsP with $\mathcal{B}(1,1)$ [9], and the EP detection [5]. The BER results of the near-optimal K -best sphere decoder (KSD) [12] are also provided.

A. Performance Evaluation

Fig. 3 shows the BER performance of the TrellisBsP detection in 8×4 , {64, 256}-QAM MIMO systems under Rayleigh channels. To ensure each detection algorithm can achieve converged results, we set a sufficient number of iterations as $Q_L = 30$ for the LAMA detection, $Q_L = 10$ for the RD-GAI-BP detection, and $Q_L = 5$ for the proposed TrellisBsP together with the BsP $\mathcal{B}(1,1)$. The damping factors for the RD-GAI-BP and the LAMA are set to $d = 0.33$ and $d = 0.5$, respectively. For the proposed TrellisBsP, the path set is constructed with $\mathcal{P}(2,1)$. For the KSD, we set $K = 32$ to ensure its near-optimal BER performance. It is observed that the proposed TrellisBsP outperforms the BsP by about 0.5 dB at BER of 10^{-3} . Compared with the EP detection, the TrellisBsP suffers

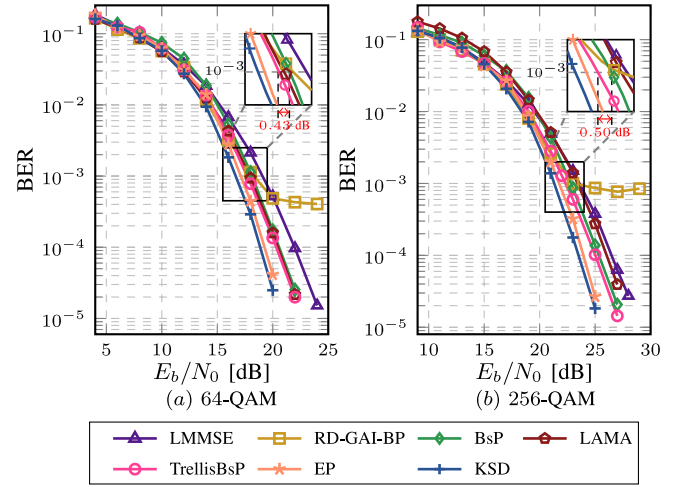


Fig. 3. BER performance of the proposed TrellisBsP detection in 8×4 MIMO systems under Rayleigh channels.

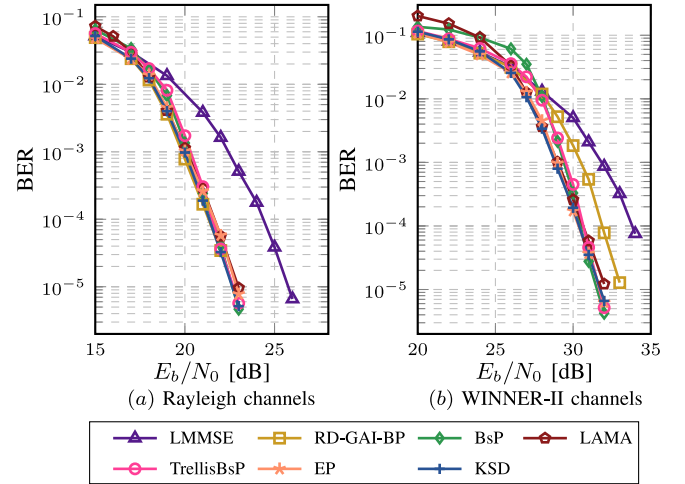


Fig. 4. BER performance of the proposed TrellisBsP detection in 128×64 , 256-QAM MIMO systems.

from about 0.55 dB performance loss in this medium-size MIMO scenario. Fig. 4 reports the BER results of the proposed TrellisBsP detection with the same parameter set as that in Fig. 3 but in 128×64 256-QAM MIMO systems under Rayleigh and WINNER-II [13] channels. It is observed that the TrellisBsP, the BsP, the LAMA, and the EP detectors all achieve near-optimal performance.

B. Computational Complexity

Table I shows the complexity comparisons of the proposed TrellisBsP, the RD-GAI-BP, the LAMA, the EP, and the BsP. Without loss of generality, we consider the proposed TrellisBsP with the path set $\mathcal{P}_i(n_m, n_c)$. First, the root path requires $2N_t$ multiplications to compute the $h_{ij}s_j$, and we note that the $h_{ij}s_j$ can be stored and reused, therefore only needing to be computed once. Then, according to Eq. (8), there are total N_p paths in this set, thus requiring to compute $N_p n_c$ additional multiplications. Recall that N_p is not a fixed number, but has an upper bound of $\binom{2N_t}{n_c} n_m n_c$. Finally, there are $2N_r$ FNs needing to update messages during each iteration, therefore the overall multiplications of the proposed TrellisBsP

TABLE I
COMPLEXITY COMPARISONS OF DIFFERENT DETECTORS

Detectors	Multiplications	Additions	Exponentiations
RD-GAI-BP [8]	$\mathcal{O}(4Q_L N_r N_t \mathcal{Q}) + \chi^\dagger$	$\mathcal{O}(4Q_L N_r N_t \mathcal{Q}) + \chi$	$\mathcal{O}(4Q_L N_r N_t \mathcal{Q})$
LAMA [3]	$\mathcal{O}(2Q_L N_t (4N_r + 4 \mathcal{Q}))$	$\mathcal{O}(2Q_L N_t (4N_r + 4 \mathcal{Q}))$	$\mathcal{O}(2Q_L N_t \mathcal{Q})$
EP [5]	$\mathcal{O}(Q_L (\frac{16}{3} N_t^3 + 8N_t \mathcal{Q}))$	$\mathcal{O}(Q_L (\frac{16}{3} N_t^3 + 8N_t \mathcal{Q}))$	$\mathcal{O}(2Q_L N_t \mathcal{Q})$
BsP [9]	$\mathcal{O}(4Q_L N_r N_t \mathcal{Q}) + \chi$	$\mathcal{O}(4Q_L N_r N_t \mathcal{Q}) + \chi$	0
TrellisBsP	$\mathcal{O}(2Q_L N_r (N_p^\dagger n_c + 2N_t)) + \chi$	$\mathcal{O}(2Q_L N_r (N_p^\dagger n_c + 2N_t)) + \chi$	0

† : $\chi = \mathcal{O}(8N_t^3)$ denotes the complexity contributed by MMSE initialization.

‡ : N_p is the number of paths and equals to $\binom{2N_t}{n_c} n_m^{n_c}$ for the worst case.

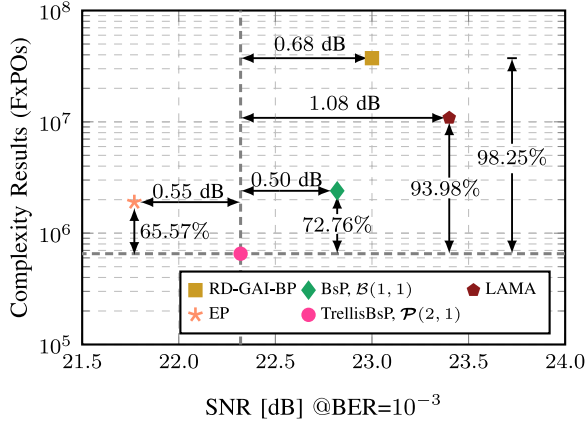


Fig. 5. Complexity comparisons of the proposed TrellisBsP and other BP detectors in an 8×4 , 256-QAM MIMO system.

are $\mathcal{O}(2Q_L N_r (N_p n_c + 2N_t) + \chi)$, where $\chi = \mathcal{O}(8N_t^3)$ is the complexity of LMMSE initialization. The expression of additions can be derived similarly. Here note that the BsP detection and the proposed TrellisBsP also require $\mathcal{O}(4Q_L N_r N_t |\mathcal{Q}|)$ and $\mathcal{O}(4Q_L N_r N_t |\mathcal{Q}| \log(n_m))$ comparisons to accomplish the sort operations, respectively. The complexity of the LAMA and EP detection are derived from [3] and [5], respectively. For more detailed comparisons, the complexity can be measured by Fixed-Point Operations (FxPOs), denoted as $\text{FxPOs} = k^2 \text{Mul.} + k \text{Add.} + mk^2 \text{Exp.}$, where k and m denote the bit width of additions and exponentiations, respectively. Fig. 5 exhibits the complexity results of the TrellisBsP with the same parameter setting as that in Fig. 3. The bit width of additions and exponentiations are set as $k = 14$ and $m = 8$ following [10], respectively. It is observed that the proposed TrellisBsP saves 93.98% and 72.76% complexity compared to the LAMA and the BsP with $\mathcal{B}(1, 1)$, respectively. Compared with the EP detection, although the proposed TrellisBsP suffers from 0.55 dB performance loss, it saves 65.57% complexity in this scenario.

V. CONCLUSION AND FUTURE WORK

A trellis-based BsP detection algorithm named TrellisBsP is proposed. By introducing the trellis representation for each FN, the proposed TrellisBsP constructs the path sets and then in parallel updates output messages associated with different paths. Compared with the EP and BsP detection, the proposed TrellisBsP demonstrates 65.57% and 72.76%

complexity reduction in the 8×4 256-QAM MIMO system, respectively. It is noted that the proposed TrellisBsP may suffer from performance degradation when applied for coded MIMO systems due to the well-known missing counter-hypothesis problem [14], which will directly be our forthcoming research endeavors.

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